

2019

COMPUTER SCIENCE

Paper : CSM-103

(Advances in Data Structure)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.**Answer question nos. 1 and 2, and any four questions from the rest.*1. Answer *any five* from the following :

2×5

- (a) State two differences between DFS and BFS, and mention their time complexities.
- (b) Prove that binary search takes $O(\log p)$ time for a list of p elements.
- (c) Show that the external path length of a 2-tree is minimum when all the leaves of the tree are on the same level or on two adjacent levels.
- (d) Define 2-3 tree and its 'largest' descendant functions. Give examples.
- (e) Critically comment : "Big-theta notation is always preferable to Big-Oh and Big-Omega".
- (f) What is the total number of binary trees with n nodes?
- (g) Can we construct a unique binary tree from Preorder and Postorder traversal if the tree is a 2-tree? Justify your answer.
- (h) What do you mean by amortized analysis of an algorithm?

2. Answer *any five* from the following :

4×5

- (a) Write down the algorithm(s) for evaluating an expression given in its infix form. Consider an example infix expression of at least six operands and four operators, and show how the algorithm(s) work.
- (b) Obtain the addressing formula for the element $A[s_1, s_2, \dots, s_n]$ in an array declared as $A[l_1 - u_1, l_2 - u_2, \dots, l_n - u_n]$. Assume a column-major representation of the array with one word per element and α the address of $A[l_1, l_2, \dots, l_n]$.
- (c) Compare with examples at least two different ways of representing graphs in a computer. Which graph representation is felt best to you and why?
- (d) Draw the following 2-3 tree that comprises a linearly ordered set, $S = \{[15, 25, 35], [55, 65, 75], [85, 95]\}$. Obtain the 2-3 tree, S' , after inserting 45 into S . Also, obtain the 2-3 tree after deleting 85 from S' .

Please Turn Over

- (e) Write an algorithm to print the elements of a singly linked list in reverse order without using any auxiliary storage.
- (f) What do you mean by balance factor of an AVL tree? Explain LR rotation with example.
- (g) What is closed hashing? What is clustering in a hash table?
- (b) Devise a simple, easy to calculate hash function for mapping three-letter words to integers between 0 and $n - 1$, inclusive. Find the values of your function on the words : ALP, MAP, PAT, SET, TAT, SAT, CTS, BAT, LAP, PAM, ATP, SIT, TAA, TAS, BAC, PAL, APM, ATA, TCS for $n = 11$. Try for as few collisions as possible.
3. (a) Construct an AVL tree following alphabetic order for the following insertion sequence : Mar, May, Nov, Aug, Apr, Jan, Dec, Jul, Feb, Jun, Oct, Sep. 7+3
- (b) State briefly how double rotation can be achieved by sequence of two single rotations.
4. Insert the keys $b_1, h_1, f_1, c_1, j_1, d_1, h_2, n_2, i_2, e_1, s_1, l_1, r_1, x_1, c_2, n_1, n_3, t_1, u_1, p_1$ into an initially empty tree, in the order given, so that the computed B-tree is of order 5. Then delete the keys one after another, from the final B-tree obtained, in the following order : h_2, r_1, p_1, d_1 and obtain subsequent B-trees. 6+4
5. (a) Define dynamic set and explain.
- (b) Consider a graph of at least ten vertices and eight edges. Process the edges one after another and compute the following :
- isolated vertices,
 - pendant vertices, and
 - the number of components belonging to the graph using disjoint set data structure.
- (c) Consider two disjoint sets of cardinality at least four each. Clearly show the node structure(s) and illustrate how these sets could be represented using linked lists. Unite the sets along with its representation, and mention the complexity of the operation. 2+4+(2+2)
6. (a) What is the utility of external sort?
- (b) Design an algorithm for any internal non-comparison sort.
- (c) Suppose you are a librarian and you have a well furnished library with a set of author name-wise physically sorted books stored in different shelves. You have purchased a new set of books. Which sorting algorithm you would use to store the entire set of books in a sorted manner physically and why? 2+6+2
7. When a sorting technique is called stable? Apply bucket sort algorithm to sort the following list of elements 5, 13, 7, 18, 35, 56, 17, 43, 89, 74. Analyze the time complexity of bucket sort. Write one disadvantage of bucket sort. 2+5+2+1
8. What do you mean by binomial heap? How does it differ from binary min heap? Explain Fibonacci heap with example. Suggest an efficient data structure to maintain a sparse matrix. 3+2+3+2